



IEDHTrans: A hybrid network with interactive encoders and differential hierarchical transformers for multi-phase breast cancer segmentation[☆]

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ABSTRACT

Breast cancer, a prevalent malignancy and leading cause of global mortality in women, requires precise tumor assessment. Although multi-phase dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) offers high sensitivity for tumor evaluation and treatment monitoring, precise primary tumor segmentation remains challenging, limiting advancements in personalized medicine. Existing segmentation methods struggle with multi-sequence DCE-MRI. Consequently, we propose IEDHTrans, a novel hybrid network leveraging multi-phase DCE-MRI information to enhance breast tumor segmentation. This network comprises an interactive encoders module for accurate multi-phase feature extraction of breast tumor features, a differential hierarchical transformer module to establish global long-distance dependencies on multi-resolution feature graphs, and a convolutional neural network decoders module for feature upsampling. Our method's effectiveness is validated through quantitative and qualitative experiments on the public MAMA-MIA dataset, the PLHN dataset, and our in-house clinical dataset. This approach consistently outperforms other advanced methods. We achieved dice coefficients of 81.22%, 77.85% and 81.83% on the MAMA-MIA, PLHN dataset and in-house clinical datasets, respectively. The source code and in-house clinical dataset are accessible at <https://github.com/WYX-gh/IEDHTrans>.

1. Introduction

Breast cancer represents a significant malignancy posing a severe threat to women's health. Compared to conventional mammography and ultrasound imaging, magnetic resonance imaging (MRI) is a highly sensitive modality for breast cancer detection, providing accurate tumor location and volume measurements from MRI images [1–4]. Notably, dynamic contrast-enhanced MRI (DCE-MRI), especially its multi-phase imaging, enhances performance in tumor detection, lesion diagnosis, and subsequent surgical treatment evaluations. The detailed temporal and spatial information from multi-phase DCE-MRI sequences, including pre- and first post-contrast phases, renders them particularly valuable for characterizing tumors. DCE-MRI stands out for its excellent contrast resolution, enabling visualization of dynamic breast tumor

changes and precise assessment of tumor behavior and response to treatment. However, manually depicting breast lesion locations on conventional MRI images is time-consuming and labor-intensive for radiologists. Therefore, developing an automatic, precise segmentation method for breast tumors is crucial for both diagnosis and medical research. In addition to streamlining clinical workflows, this approach enhances the reliability and consistency of tumor assessments, ultimately contributing to more effective personalized treatment strategies.

Deep learning has recently driven significant advancements in medical image segmentation [5–7]. Convolutional neural networks (CNNs) (e.g., the prevalent 3D-UNet architecture [8]) and transformer-based models (e.g., the Swin-Transformer [9]) have been crucial in enhancing accuracy. Notably, the U-Net network excels at capturing global

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features while retaining detailed information via its encoder–decoder architecture and skip connections. Meanwhile, transformer models excel in processing global contextual information, ideal for complex spatial relationships [10]. To maximize the strengths of both CNNs and transformers, researchers have proposed hybrid networks that integrate these architectures. For instance, TransUNet [11] integrates transformer with U-Net to capture global encoder information before upsampling for segmentation. Another example is TransResNet [12], which effectively integrates local and global features via parallel CNN and transformer encoder branches, augmented by cross-graft modules (CGM), to improve high-resolution medical image segmentation. Furthermore, ConvFormer [13] introduces a novel CNN-style transformer designed to mitigate attention collapse in medical image segmentation by incorporating pooling, CNN-style self-attention, and convolutional feed-forward networks.

Hybrid approaches capitalize on CNNs' local feature extraction and transformers' global contextual understanding, enhancing the accuracy and efficiency of medical image segmentation. This combination allows models to better capture complex tumor characteristics, ultimately contributing to more precise diagnoses and personalized treatments.

However, these methods are predominantly designed for single-modal inputs and often struggle to process multi-phase data effectively. Concatenating multi-phase data directly as a single-modal input fails to adequately model the intricate inter-sequence relationships and complementary information across different imaging sequences. This limitation substantially inhibits their capability to achieve robust segmentation in multi-phase DCE-MRI scenarios, especially when handling heterogeneous tissue characteristics or dynamic contrast enhancement patterns. Specifically, directly concatenating multiphase DCE-MRI images along the channel dimension gives rise to three major structural issues: (1) Feature explosion and redundancy: As the number of phases increases, the input channels grow linearly, leading to a substantial expansion of parameters in the early convolutional layers. This not only results in excessive memory consumption but also introduces a large amount of redundant information, thereby weakening the model's ability to effectively represent the core tumor region. (2) Dynamic enhancement information loss: Treating images from different time points as independent channels disrupts the intrinsic temporal evolution patterns of DCE-MRI, making it challenging for the model to capture morphological differences in enhancement curves between tumors and normal tissues. (3) Lack of cross-phase interaction: During the convolution of concatenated feature maps, interactions are limited to implicit operations within local receptive fields, without an explicit mechanism for cross-phase information alignment. Consequently, the model fails to enforce the constraint of "temporal consistency across the same spatial position at different phases", ultimately impairing the segmentation accuracy for tumors with indistinct boundaries or heterogeneous structures.

Addressing the limitations of conventional multi-phase DCE-MRI segmentation methods, we propose a novel hybrid encoder–decoder architecture. This optimized framework, adapted for multi-phase medical image analysis, strategically integrates 3D CNNs and hierarchical transformers, and features two key components: (1) An interactive encoder (IE) module, which employs a dual-branch 3D CNN to specifically extract and interact with corresponding features from different phases; and (2) A differential hierarchical transformer (DHFormer) module, designed to model global contextual relationships across sequences. Although CNNs excel at capturing fine-grained local details, they are inherently limited in modeling long-range dependencies. Conversely, transformers effectively leverage self-attention for long-range dependencies and global semantics. Integrating both architectures, the proposed method achieves a more comprehensive and context-aware representation of multi-phase medical images. Specifically, we first utilize the most prevalent 3D medical image segmentation network, nnUNet [14] as the data preprocessing network. Second, we designed a parallel 3D CNN encoder with multi-stage feature interaction, followed

closely by a DHFormer module [15,16] to capture the global dependencies between bottleneck features. Finally, we employed a 3D CNN decoder to generate corresponding prediction masks. The experimental results on the publicly available MAMA-MIA [17] dataset, the 100-case dataset publicly available by PLHN [18], and an internal dataset demonstrated that the proposed method achieves accurate breast tumor segmentation and consistently outperforms state-of-the-art approaches.

Generally, the main contributions of this study are summarized as follows: (1) We designed an IE module to extract features from pre- and first post-contrast images, enabling information interaction among multi-phase image. This integrates complementary features from different stages for accurate, multi-stage breast tumor feature extraction. (2) DHFormer was designed to establish global long-range dependencies on feature graphs at different resolutions, facilitating multi-scale volume feature computation. (3) Experiments on both the open MAMA-MIA dataset, PLHN dataset and our in-house clinical dataset demonstrated that the proposed method outperforms other conventional breast tumor segmentation approaches. (4) We developed and published a novel DCE-MRI breast dataset to enable further research and evaluation of the proposed method.

The remainder of this paper is organized as follows. Section 2 briefly outlines the most relevant studies in breast tumor segmentation. Section 3 comprehensively describes the proposed approach. Section 4 presents the extensive experiments conducted and discusses the obtained results. Finally, Section 5 concludes the paper.

2. Related works

2.1. CNN-based segmentation methods

CNN-based segmentation methods are crucial in medical image segmentation. For example, MHL [19] effectively addressed class imbalance and background interference in tumor segmentation via multi-level feature extraction and full convolutional up/downsampling. V-Net [20] enhanced 3D medical image segmentation accuracy by extracting 3D volumetric features using a 3D convolutional network. DMFNet [21] achieved high-precision tumor segmentation by employing multiple lightweight 3D convolutional networks and 3D expanded convolutions to reduce computational costs and build multi-scale features. CE-Net [22] introduced an architecture with a feature encoder, a context extractor, and a feature decoder. Its context extractor uniquely employs dense hollow convolutional blocks and residual multi-kernel pooling blocks to effectively capture advanced and spatial information. However, despite these advancements, CNN-based methods struggle with capturing global information.

2.2. Transformer-based segmentation methods

Transformer-based segmentation methods are also crucial in medical image segmentation. For example, Dual-ViT [23] proposed a novel transformer architecture, dual-View Transformer, comprising semantic paths for global semantics (reducing complexity) and pixel paths for learning details. MISSFormer [24] enhanced feature discrimination and context modeling via ReMix-FFN and ReMixed Transformer Context Bridge. MAXFormer [25] excels at capturing long-range dependencies in high-resolution medical images, using local–global and external attention, while reducing spatial detail loss due to downsampling with a refined fusion module. However, a common limitation is that transformer-based methods typically require extensive training data to achieve superior segmentation performance. NestedFormer [26], built upon a multi-encoder–single-decoder Transformer framework, explicitly characterized both intra- and inter-modal relationships of multi-modal MRI for the first time. By incorporating the NMaFA module to fuse high-level features and employing MSG gating to refine skip connections, the method demonstrated remarkable performance in brain tumor segmentation.

2.3. Hybrid models for segmentation

2.3.1. CNN-transformer hybrids

Integrating CNNs and transformers, hybrid segmentation models exhibit excellent medical field performance. For instance, AST-Net [27] uses a lightweight convolution module for local features and an axial-spatial converter module that learns long-distance dependencies through axial and spatial attention, capturing multi-view and multi-scale features. COTR [28], another example, employs deformable transformers to capture local and global prior information, enabling the model to utilize the most relevant priors for image regions. Networks form multi-scale pipelines via recursive scaling during reasoning, yielding high-precision corresponding points. LW-CTrans [29] strategically combined the advantages of CNN and transformer in different phases to efficiently segment small targets in 3D medical image segmentation.

2.3.2. Other hybrid methods

Beyond conventional CNN-Transformer hybrid architectures, several recent studies have explored alternative paradigms for modeling long-range dependencies. For instance, Serp-Mamba [30] introduces a selective state-space mechanism for high-resolution retinal vessel segmentation, achieving linear complexity while maintaining a global receptive field. SegMamba [31] and its extended version SegMamba-V2 [32] extend Mamba-style sequential modeling to 3D medical volumes, delivering competitive accuracy while alleviating the quadratic computational burden of self-attention. Diff-UNet [33] progressively refines diffusion models through Multi-granularity Boundary Aggregation, Monte Carlo Diffusion, and a Progressive Uncertainty-driven Refinement strategy, and represents the first effort to transform diffusion models into universal 3D segmentation tools, substantially outperforming existing methods across multiple tasks. Hybrid-MIM [34] proposes a two-stage hierarchical masking strategy that enables efficient self-supervised pretraining at the pixel, region, and sample semantic levels. Moreover, it flexibly supports both CNN and Transformer encoders/decoders, achieving superior accuracy and efficiency in a wide range of segmentation tasks.

2.4. Breast cancer segmentation

Recently, deep learning techniques are prevalent and have achieved remarkable results in the field of breast cancer segmentation [35,36]. For example, PLHN [18] proposed a prototype learning-guided hybrid network [37,38] that effectively captures the global dependence of tumor features by combining convolutional, inverse convolutional, and transformer layers. DSTC-Net [39] introduced the DST-C network to address annotation scarcity via self-supervised learning. HAU-Net [40], by introducing L-G transformer blocks and cross-attention modules, leveraged transformer's long-range dependencies and CNN's local detail representation.

2.5. Feature interaction

Feature interaction is pivotal in multi-phase medical image segmentation. For instance, TransFusion [41] achieved MRI-CT feature interaction using a cross-modal attention mechanism. I2U-Net [42] enhances historical information utilization and addresses conventional U-shaped network limitations by modeling cross-path and cross-layer information interaction. Another approach [43], extracted features from each mode via a dual-stream architecture, and then utilized a cross-transformer to query and retrieve responses between modes, enabling effective dual-mode feature interaction.

2.6. Cross-modal joint modeling

Recent studies have investigated more robust joint modeling of heterogeneous MRI phases and modalities. XLSTM-HVED [44] integrates a Vision-XLSTM reconstruction branch with a heteromodal variational encoder-decoder, enabling simultaneous missing-sequence imputation and brain tumor segmentation. SCUNet++ [45] further advances the fusion paradigm by combining Swin-UNet and CNN bottlenecks through multi-fusion dense skip connections, thereby facilitating enriched cross-phase information exchange. Despite their strong performance, both approaches introduce substantial computational overhead: XLSTM-HVED due to bidirectional long-range recurrence, and SCUNet++ owing to densely rewired skip pathways. These limitations reduce their practicality in resource-constrained clinical settings.

We propose a lightweight hybrid segmentation network, termed IEDHTrans, whose core objective is to achieve both complementary multi-phase information interaction and global multi-scale dependency modeling within a unified encoder-decoder framework, while introducing only minimal parameter overhead. Specifically, the network begins with a parallel dual-stream encoding subnet and integrates pre- and first post-contrast features via a concise yet efficient Multi-Phase Feature Interaction (MPFI) block during the encoding stage. This design preserves intertemporal complementary information while avoiding the substantial computational burden typically incurred by cross-modal reconstruction or dense skip connections commonly used in conventional approaches. The fused bottleneck features are then processed by DHFormer, a Transformer module specifically designed for multi-scale long-range dependency modeling, thereby facilitating faster convergence through enhanced global context capture. Overall, IEDHTrans achieves precise segmentation of multi-phase breast DCE-MRI while preserving the lightweight nature of the network.

3. Method

This section describes the proposed IEDHTrans model. We begin by outlining its overall architecture, then detail the IE, DHFormer, and decoder modules. Finally, we explain the loss function used for training.

3.1. IEDHTrans architecture

Given a set of multi-phase breast cancer images, including pre- and first post-contrast images, we attempt to accurately segment the shared tumor regions, which is achieved by the IEDHTrans co-segmentation framework. Fig. 1 illustrates the overall IEDHTrans architecture, where we designed an encoder-decoder architecture with skip connections [46] to develop the network backbone. To effectively handle the multi-phase data, we construct the encoder with an IE module and a DHFormer module, including the decoder relative to the encoder. The IE module leverages two independent 3D residual convolutional networks to process pre- and first post-contrast images separately, capturing their specific features. Meanwhile, complementary features from images at different phases in the model are integrated to improve segmentation accuracy. The DHFormer module is designed to address CNNs' limitation in modeling long-range pixel relationships. By establishing global dependencies between feature maps at different resolutions, it improves segmentation accuracy. The decoder module comprising 3D deconvolution operations progressively restores the spatial resolution of feature maps while maintaining structural integrity. It mirrors the encoder structure, forming an encoder-decoder framework optimized for 3D feature reconstruction. To optimize segmentation performance, we employ a combined cross-entropy and Dice loss as the objective function. Cross-entropy loss optimizes pixel-wise classification accuracy, while Dice loss improves region-based overlap between predictions and ground truth, addressing class imbalance issues. The detailed structures of the IE, DHFormer, decoder, and loss function modules are presented in the following four subsections.

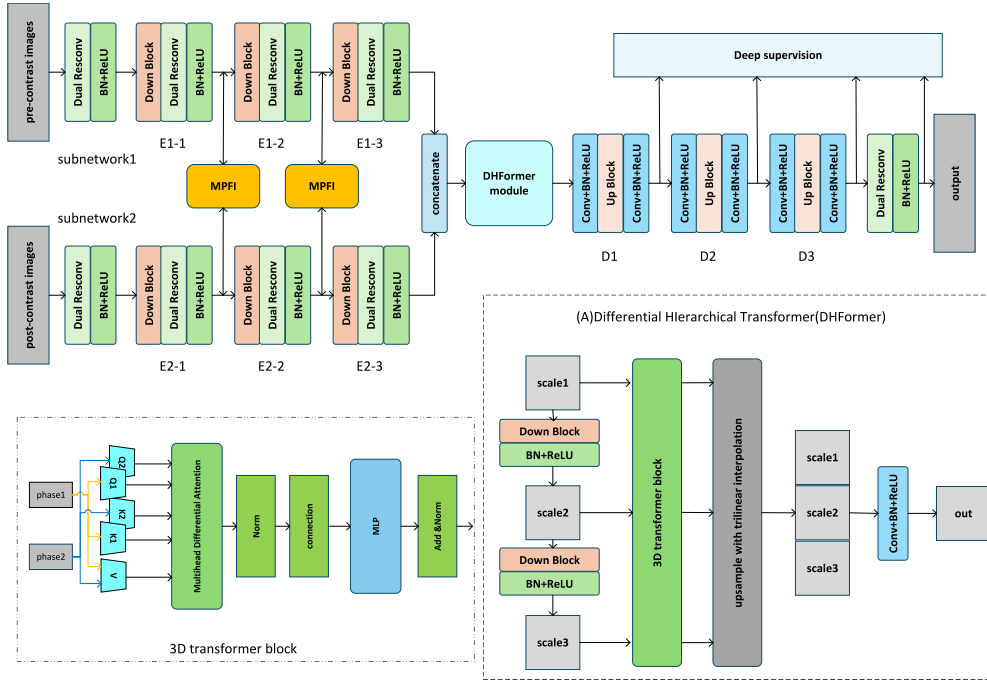


Fig. 1. The proposed IEDHTrans model includes an interactive encoder (IE) with two 3D residual convolutional networks, a differential hierarchical transformer (DHFormer) module, and a decoder. The IE captures features from pre- and first post-contrast images using subnetwork 1 and subnetwork 2, and fuses complementary features at each stage using the MPFI module. The DHFormer establishes global dependencies across multi-scale resolutions using 3D transformer blocks with multi-head differential attention (MHDA) and upsampling. The decoder restores resolution through an upsampled convolutional block with skip connections and deep supervision that comes with nnUnet. (A) Differential Hierarchical Transformer (DHFormer).

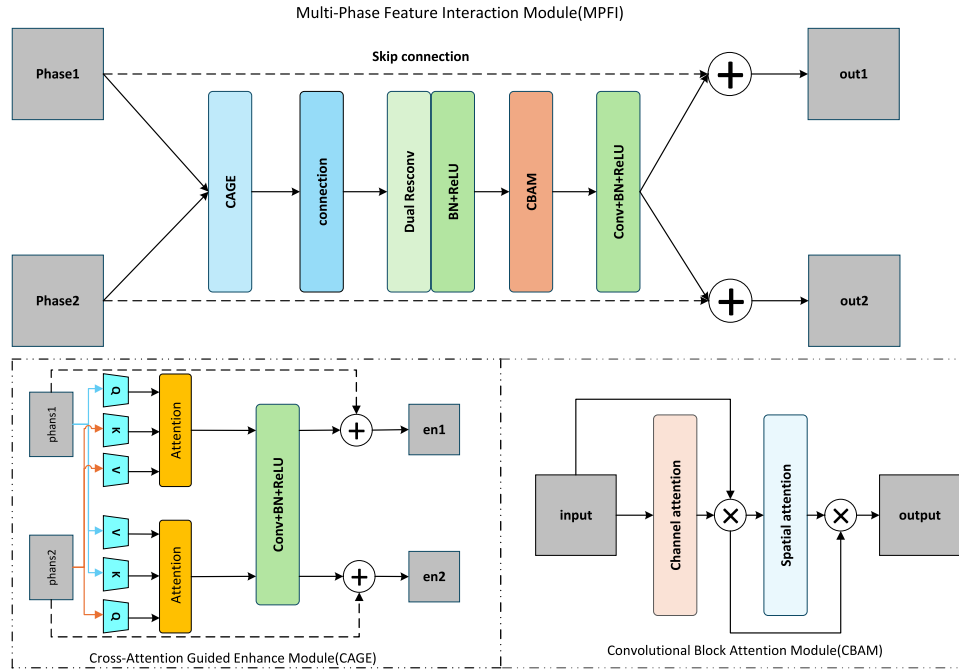


Fig. 2. The structure diagram of the MPFI module. The MPFI Module mainly relies on Cross-Attention Guided Enhance Module(CAGE) and the Convolutional Block Attention Module (CBAM) to refine the channel-space dual attention of the interaction features.

3.2. IE module

In contrast to XLSTM-HVED, which relies on full-sequence reconstruction for segmentation regularization, and SCUNet++, which requires dense skip connections and re-routing at every resolution, our proposed IE module is designed with an efficiency-first principle. It effectively extracts and fuses local features from multiphase data, while

employing two independent encoders to process images from different modalities. This design enables the capture of modality-specific representations and enhances segmentation performance through feature fusion. Meanwhile, feature interaction is conducted utilizing the MPFI module between these two independent encoders to integrate the complementary features in the images at different stages of the model. As illustrated in Fig. 1, the parallel CNN encoder module comprises two

identical 3D residual convolutional networks. The size of the image input to the two CNN encoders is $D \times H \times W$, where D , H , and W denote the depth, height, and width of the input image, respectively. The 3D residual convolution block is the basic component of this network, comprising a batch normalization layer [47], leaky rectified linear unit (ReLU) [48] activation function, and convolutional layer. The residual convolution operation can be expressed as:

$$\text{ConvBlock} = \sigma \left(\text{BN} \left(\sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^2 X_{D+i} Y_{D+j} Z_{D+k} \cdot W_{i,j,k} \right) \right) \quad (1)$$

$$A = \text{ConvBlock}(X, W_1) \quad (2)$$

$$F_{\text{out}} = X + \text{ConvBlock}(A, W_2) \quad (3)$$

where X , W , BN , σ , A , and F_{out} denote the input feature map, convolutional kernel, batch normalization, leaky ReLU activation function, intermediate feature map, and output feature map, respectively.

In Fig. 1, subnetwork-1 and subnetwork-2 both comprise three residual convolution modules. Each residual convolution module comprises one residual convolution blocks and one downsampling convolution block. Each residual convolution block performs two residual convolution operations, using a $3 \times 3 \times 3$ convolution kernel with an edge padding is 1. Correspondingly, intermediate features E1-1, E1-2, E2-1, and E2-2 are extracted from the two encoder subnetworks and relayed to the decoder via skip connections. Notably, the first downsampled convolution block's stride in the 3D residual convolution module used for E1-1 and E1-2 is (1, 2, 2), i.e., the x - and y -axes are downsampled while the z -axis is not, resulting in an output size of $D \times H/2 \times W/2$. Subsequent 3D residual convolution modules in E1-2, E1-3, E2-2, and E2-3 have a stride of (2, 2, 2), downsampling all x -, y -, and z -axes. Their output sizes are $D/2 \times H/4 \times W/4$ and $D/4 \times H/8 \times W/8$ for E1-2 and E2-2 and E1-3 and E2-3, respectively. The final bottleneck features of the two encoders are fused and inputted into the DHFormer module, and their fused bottleneck feature size is $D/4 \times H/8 \times W/8$.

Meanwhile, the MPFI module integrates complementary features from multi-phase images, enabling their interaction and fusion within the module.

As illustrated in Fig. 2, the module first facilitates image enhancement through the CAGE block, after which the enhanced features from the two stages are combined and further refined via a residual convolution operation. To improve the expressiveness of the fused representations, CBAM [49] is subsequently employed to account for both channel and spatial information of the feature maps. The CAGE block serves as a cross-attention mechanism specifically designed for guiding feature enhancement, which can be formulated as:

$$\text{out} = \text{LayerNorm}(Q \cdot K^T \cdot V) + Q \quad (4)$$

where LayerNorm denotes normalization, Q corresponds to the features of phase 1 with dimensions $B \times C_Q \times D \times H \times W$, while K and V denote the features of phase 2 with dimensions $B \times C_K \times D \times H \times W$ and $B \times C_V \times D \times H \times W$, respectively.

As shown in Fig. 2, the cross-attention is applied twice. In the first instance, phase 1 serves as Q while phase 2 provides K and V , thereby enhancing the feature representation of phase 1 with information from phase 2. Similarly, the second application uses phase 1 to enhance the features of phase 2. The enhanced features are then combined with the original phase via residual connection before proceeding to subsequent operations.

The CBAM module is a channel and spatial attention mechanism specifically designed for 3D feature data. The core idea is to dynamically adjust the feature response intensity of each channel and space by learning the dependency relationships between channels and spaces. Specifically, the CBAM module contains two consecutive operations: channel and spatial attention.

The channel attention can be expressed as:

$$W_{\text{channel}} = \sigma(\text{Conv3d}_2(\text{ReLU}(\text{Conv3d}_1(\text{AdaptiveAvgPool3d}(x)))) \quad (5)$$

where σ , Conv3d_1 , AdaptiveAvgPool3d and Conv3d_2 represent the Sigmoid activation function, convolutional layer reducing the number of channels from C to 1, instance normalization operation [47], and convolutional layer maintaining the number of channels as 1, respectively.

The spatial attention can be expressed as:

$$W_{\text{spatial}} = \sigma(\text{Conv3d}_4(\text{ReLU}(\text{InstanceNorm3d}(\text{Conv3d}_3(x)))) \quad (6)$$

where σ , Conv3d_3 , InstanceNorm3d and Conv3d_4 denote the Sigmoid activation function, convolutional layer reducing the number of channels from C to 1, instance normalization operation [47], and convolutional layer maintain the number of channels as 1, respectively.

The final output after the MPFI module:

$$y = x \cdot W_{\text{channel}} \cdot W_{\text{spatial}} \quad (7)$$

where x , W_{channel} , W_{spatial} the input feature, channel attention weight, spatial attention weight (element-wise multiplication operation), respectively.

After passing through the aforementioned attention module, the fused features will further undergo a convolution operation with a convolution kernel set to "1". Subsequently, the features are summed with the input features, passed into the corresponding phase, and fused with the features of the original phase. Specifically, the proposed MPFI module conducted feature interactions between E1-1 and E2-1, including E1-2 and E2-2, and transferred the interacted features to E1-2 and E2-2 and E1-3 and E2-3. Finally, the image obtained after interaction is fused with the original image and passed to the next layer as a new input.

3.3. DHFormer module

To address the limitations of conventional CNN in modeling long-range pixel relationships, we designed the DHFormer module to better capture multi-scale features by establishing global dependencies between features of different resolutions. Unlike the general Differential Transformer [16], the differential attention mechanism in DHFormer is specifically tailored for modeling dynamic contrast differences in multi-phase DCE-MRI. As illustrated in Fig. 1 (A), DHFormer treats the features at the same spatial position across different phases (pre- and first post-contrast) as two independent signal sources. By explicitly computing the difference between their attention maps, it amplifies the subtle variations between pre- and post-contrast phases. These amplified differences constitute the most critical and clinically significant temporal feature of DCE-MRI. In addition to improving the segmentation accuracy, the proposed DHFormer can accelerate the convergence of the network via multi-scale feature fusion.

Fig. 1 (B) shows that the DHFormer module's input is a fusion of the output of two parallel CNN encoders with a size of $D/4 \times H/8 \times W/8$. In the DHFormer module, we first sample the input features for two convolution systems, then perform linear transformations for the input features and features after two down-samples, and then adopt the 3D transformer layer in turn. The 3D transformer layer primarily comprises multi-head differential attention (MHDA) and MLP. Meanwhile, to reduce the challenge posed by the transformer's video memory occupation, we also combined the flash attention [50] to reduce video memory usage. The implementation method of the differential attention module is expressed as:

$$\text{MHDA}(X) = \text{Concat}(\overline{\text{head}}_1, \overline{\text{head}}_2, \dots, \overline{\text{head}}_h) W_O \quad (8)$$

where each head is computed as:

$$\text{head}_i = \text{DiffAttn}(X; W_{Q_i}, W_{K_i}, W_{V_i}, \lambda) \quad (9)$$

$$\overline{\text{head}}_i = (1 - \lambda_{\text{init}}) \cdot \text{LN}(\text{head}_i) \quad (10)$$

where $\text{LN}(\cdot)$: RMSNorm [51] or GroupNorm [52] applied to each head, $(1 - \lambda_{\text{init}})$ fixed scaling required to align gradient flow with standard transformer. $W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ is a learnable projection matrix.

The differential attention mechanism DiffAttn is given by:

$$[Q_1; Q_2] = XW_{Q_1}, \quad [K_1; K_2] = XW_{K_1}, \quad V = XW_{V_1} \\ \text{DiffAttn}(X) = \left(\text{softmax}\left(\frac{Q_1 K_1^T}{\sqrt{d}}\right) - \lambda \cdot \text{softmax}\left(\frac{Q_2 K_2^T}{\sqrt{d}}\right) \right) V \quad (11)$$

where $X \in \mathbb{R}^{N \times d_{\text{model}}}$ denotes the input feature vector. Q_1 and K_1 are obtained by applying learnable linear projections W_{Q_1} and W_{K_1} to the pre-contrast feature map, thereby encoding global query-key relationships of the non-enhanced signals. Similarly, Q_2 and K_2 are generated from the first post-contrast feature map via projections W_{Q_2} and W_{K_2} , capturing early-enhancement information. The value matrix V is constructed by concatenating both pre- and first post-contrast features along the channel dimension, followed by a linear projection through W_V , thereby preserving comprehensive spatial and temporal information from both phases. A learnable scalar λ is introduced to regulate the intensity of suppression.

The learnable scalar λ is computed as:

$$\lambda = \exp(\lambda_{q1} \cdot \lambda_{k1}) - \exp(\lambda_{q2} \cdot \lambda_{k2}) + \lambda_{\text{init}} \quad (12)$$

Here, $X \in \mathbb{R}^{N \times d_{\text{model}}}$ is the input feature vector, $\lambda \in \mathbb{R}$ is a learnable scalar, $\lambda_{q1}, \lambda_{k1}, \lambda_{q2}, \lambda_{k2} \in \mathbb{R}^d$ are learnable vectors associated with the corresponding phase, and $\lambda_{\text{init}} \in (0, 1)$ is an initialization constant. As indicated in the original Differential Transformer study [16], the following initialization strategy:

$$\lambda_{\text{init}} = 0.8 - 0.6 \exp(-0.3(l - 1))$$

performs well in practice, where $l \in [1, L]$ denotes the layer index. This formulation ensures adaptive enhancement of differential features across layers.

This design enables λ to adaptively amplify or suppress the relative contribution of the first post-contrast phase to the pre-contrast phase during training, thereby improving the characterization of enhancement variations across different lesions.

Scale1's output size the transformer layer is $D/4 \times H/8 \times W/8$; Scale2's output size after the transformer layer is $D/8 \times H/16 \times W/16$, and scale3's output size after the transformer layer is $D/16 \times H/32 \times W/32$. Finally, the results of the three transformers are up-sampled to $D/4 \times H/8 \times W/8$ and fused via convolution with a convolution kernel of 1. The fused output size is $D/4 \times H/8 \times W/8$. The feature map is fed as an input to the decoder and fused with the feature maps of the CNN encoder, E1-3 and E2-3, and then up-sampled.

We observe that the differential attention mechanism generates attention weights by computing the difference between two softmax outputs, which seemingly violates the non-negativity constraint of conventional attention weights. This design, however, is inspired by the recently proposed Differential Transformer [16] in natural language processing, whose core idea is to suppress context-irrelevant attention noise through differential operations, thereby strengthening the emphasis on salient information. Mathematically, a negative differential attention weight does not represent an invalid weight but rather functions to attenuate irrelevant regions. This mechanism operates analogously to a differential amplifier in signal processing, where common-mode noise is eliminated via subtractive operations.

3.4. Decoder module

To progressively restore spatial resolution while maintaining structural integrity in volumetric data processing, we propose a hierarchical decoder architecture employing parametrized transposed convolution

operations. The decoder exhibits structural symmetry with dual encoder sub-networks, forming an encoder-decoder framework optimized for 3D feature reconstruction, as illustrated in Fig. 1.

The decoder receives output features from the DHFormer module and processes them through several 3D deconvolution modules. Each module is carefully designed to balance computational efficiency and reconstruction quality, progressively enhancing the resolution of the feature maps across multiple stages (D1, D2, and D3). Each 3D deconvolution module within the decoder integrates a carefully orchestrated sequence of operations. The process begins with a transposed convolutional layer that performs adaptive up-sampling using learnable parameters, which ultimately enables context-aware feature expansion optimized for input data characteristics. Subsequently, the refinement convolutional layer employing a $3 \times 3 \times 3$ kernel with stride 1 is adopted to enhance the up-sampled features and mitigate potential artifacts introduced during expansion. Next, batch normalization stabilizes the training process by normalizing feature distributions, facilitating faster convergence and improved gradient flow. The sequence concludes with a leaky ReLU activation that introduces non-linearity while preserving gradient information during backpropagation, thus enhancing the representational capacity and expressiveness of the feature maps. This architecture enables effective reconstruction of volumetric data while maintaining spatial coherence throughout the decoding process. In the first two decoder stages (D1 and D2), the transposed convolution layers employ a stride of 2 along all spatial axes (x, y, and z), doubling the feature-map resolution at each stage. Consequently, D1 and D2's outputs have dimensions $D/2 \times H/4 \times W/4$ and $D \times H/2 \times W/2$, respectively. In the final decoder stage (D3), the transposed convolution uses a stride of (1, 2, 2), up-sampling along the x- and y-axes while preserving the resolution along the z-axis. Consequently, D3's output has dimensions $D \times H \times W$, fully restoring the original volumetric resolution.

3.5. Loss function

We utilize a combined cross-entropy and Dice loss [53] for our segmentation loss function. Cross-entropy optimizes pixel-wise classification accuracy, while Dice loss enhances region-based overlap and addresses class imbalance issues. To maximize the model's multi-resolution output, we implement a deep supervision strategy. This approach integrates additional loss terms at intermediate network layers, enabling simultaneous optimization of features across multiple scales, thus improving model accuracy and robustness.

Assume the network comprises n outputs with different resolutions, with each output corresponding to a loss L_i and a weight w_i . The deep supervised loss L_{total} can be expressed as:

$$L_{\text{total}} = \sum_{i=1}^n w_i \left(L_{\text{Dice}}^{(i)} + \lambda L_{\text{CE}}^{(i)} \right) \quad (13)$$

where $L_{\text{Dice}}^{(i)}$ is the Dice loss in layer i ; $L_{\text{CE}}^{(i)}$ is the cross-entropy loss in layer i ; λ is the equilibrium parameter; w_i is the weight of the loss in layer i . Notably, the weight decreases exponentially with decreasing resolution

Each deeply supervised output Dice loss can be expressed as:

$$L_{\text{Dice}}^{(i)} = 1 - \frac{2 \sum_{p \in P^{(i)}} \text{Softmax}(O_i^{(p)}) \cdot G_i^{(p)}}{\sum_{p \in P^{(i)}} \text{Softmax}(O_i^{(p)}) + \sum_{p \in P^{(i)}} G_i^{(p)} + \epsilon} \quad (14)$$

where $O_i^{(p)}$ is the predicted segmentation result of the i th layer output; $G_i^{(p)}$ is the one-hot encoding of the ground truth segmentation; $P^{(i)}$ is the set of pixels in the i th layer; $\text{Softmax}(O_i^{(p)})$ is the probability distribution of the predicted segmentation; ϵ is a smoothing term to avoid division by zero.

$$L_{\text{CE}}^{(i)} = -\frac{1}{|P^{(i)}|} \sum_{p \in P^{(i)}} \sum_{k=1}^K G_i^{(p,k)} \log \left(\text{Softmax}(O_i^{(p,k)}) \right) \quad (15)$$

Table 1

Comparison of IEDHTrans on MAMA-MIA datasets with other methods. The values in bold and underlined values represent the optimal and second-best values, respectively. The data in this table is a combination of mean $\pm$ standard deviation.

	DPC(%) $\uparrow$	ASSD(mm) $\downarrow$	VOE(%) $\downarrow$	HD95(mm) $\downarrow$	Sensitivity(%) $\uparrow$	Specificity(%) $\uparrow$
nnFormer	67.82 $\pm$ 23.09	28.14 $\pm$ 36.71	44.46 $\pm$ 24.32	89.36 $\pm$ 86.59	82.93 $\pm$ 16.68	99.89 $\pm$ 0.17
PA-Net	75.13 $\pm$ 20.79	17.16 $\pm$ 29.86	35.97 $\pm$ 23.44	58.99 $\pm$ 78.34	82.59 $\pm$ 18.18	99.95 $\pm$ 0.07
PHTrans	75.23 $\pm$ 21.17	20.99 $\pm$ 35.10	35.78 $\pm$ 23.43	64.41 $\pm$ 84.66	85.71 $\pm$ 15.81	99.91 $\pm$ 0.26
nnUNet	76.08 $\pm$ 20.47	17.02 $\pm$ 30.23	34.81 $\pm$ 23.27	57.97 $\pm$ 80.04	83.47 $\pm$ 17.17	99.93 $\pm$ 0.19
PLHN	74.55 $\pm$ 19.65	14.37 $\pm$ 24.66	37.14 $\pm$ 22.14	55.35 $\pm$ 74.32	78.55 $\pm$ 21.20	99.94 $\pm$ 0.11
STU-Net	77.10 $\pm$ 19.74	<u>14.30 $\pm$ 27.82</u>	<u>33.69 $\pm$ 22.52</u>	<u>50.21 $\pm$ 75.42</u>	84.21 $\pm$ 17.51	99.94 $\pm$ 0.13
IEDHTrans	81.22 $\pm$ 16.75	6.99 $\pm$ 15.06	29.06 $\pm$ 20.14	30.39 $\pm$ 58.34	<u>84.64 $\pm$ 18.03</u>	99.96 $\pm$ 0.08

Table 2

Comparison of IEDHTrans on PLHN datasets with other methods. The values in bold and underlined values represent the optimal and second-best values, respectively. The data in this table is a combination of mean $\pm$ standard deviation.

	DPC(%) $\uparrow$	ASSD(mm) $\downarrow$	VOE(%) $\downarrow$	HD95(mm) $\downarrow$	Sensitivity(%) $\uparrow$	Specificity(%) $\uparrow$
nnFormer	63.30 $\pm$ 21.25	36.59 $\pm$ 31.08	50.30 $\pm$ 22.28	122.07 $\pm$ 60.46	86.80 $\pm$ 10.42	99.91 $\pm$ 0.10
PA-Net	67.80 $\pm$ 19.25	32.35 $\pm$ 30.13	45.72 $\pm$ 20.99	114.97 $\pm$ 67.91	87.24 $\pm$ 11.62	99.94 $\pm$ 0.06
PHTrans	67.37 $\pm$ 21.97	31.53 $\pm$ 32.55	45.36 $\pm$ 23.61	101.56 $\pm$ 71.71	90.89 $\pm$ 9.44	99.92 $\pm$ 0.10
nnUNet	71.27 $\pm$ 20.62	24.91 $\pm$ 28.64	41.04 $\pm$ 22.79	82.51 $\pm$ 71.01	87.58 $\pm$ 11.57	99.95 $\pm$ 0.06
PLHN	64.64 $\pm$ 21.92	31.30 $\pm$ 26.81	48.89 $\pm$ 21.21	107.84 $\pm$ 55.66	78.08 $\pm$ 23.67	99.95 $\pm$ 0.04
STU-Net	73.04 $\pm$ 20.00	<u>20.80 $\pm$ 26.76</u>	<u>39.08 $\pm$ 21.95</u>	<u>74.02 $\pm$ 69.11</u>	87.26 $\pm$ 12.63	99.96 $\pm$ 0.04
IEDHTrans	77.85 $\pm$ 17.66	12.25 $\pm$ 18.17	33.41 $\pm$ 20.23	52.15 $\pm$ 64.63	<u>87.63 $\pm$ 13.46</u>	99.97 $\pm$ 0.04

where K is the number of classes; $G_i^{(p,k)}$ is the ground truth label of the p th pixel in the i th layer belonging to the k th class; $\text{Softmax}(O_i^{(p,k)})$ is the predicted probability of the p th pixel in the i th layer belonging to the k th class; $|P^{(i)}|$ is the number of pixels in the i th layer.

4. Experiments

4.1. Dataset and implementation details

The MAMA-MIA dataset employed in this study was assembled from The Cancer Imaging Archive for pre-treatment DCE-MRI cases, totaling 1506 cases collected from four centers. Here, 16 experts manually segmented primary tumors in these cases. The dataset was randomly partitioned into a training set (964 patients), a validation set (242 patients), and a test set (300 patients). In addition to the publicly available MAMA-MIA dataset, we also used 100 patients publicly available on PLHN [18] as an external test set to verify the effectiveness of our method. To further assess the model's generalizability, an independent dataset of 31 cases was collected from the First Affiliated Hospital of Shandong First Medical University. Experienced clinicians manually annotated these cases utilizing ITK-SNAP, which helped assess the model's robustness and clinical applicability.

To improve the network's segmentation efficiency and accuracy, we employed the nnUNet data preprocessing module to crop the input image's non-zero region, resample, and normalize, unifying the voxel space size of different data. For each modality, both the image and the segmentation annotation image were cropped to their non-zero region to reduce the image size. Using interpolation and other methods, each mode's image and segmented image are resampled to ensure a consistent size in the voxel space. Finally, normalization is performed on the image.

The proposed hybrid network was implemented using PyTorch [54] on an Nvidia GeForce RTX 4090 GPU equipped with 24 GB of RAM. The network was trained using an Adam optimizer with an initial learning rate of 0.01. Prior to processing, each image was segmented into small patches with dimensions of $64 \times 192 \times 192$, and a batch size of 2 was adopted. The training iterated through the default 1000 epochs of nnUNet.

4.2. Evaluation metrics

To quantitatively assess the proposed network's performance, we employed seven prevalent metrics for segmentation evaluation. Specifically, we measured the segmentation results using the case-by-case Dice coefficient (DPC), average symmetric surface distance (ASSD), volume overlap error (VOE), Hausdorff95 distance (HD95), sensitivity, and specificity. These metrics collectively capture both overlap accuracy and boundary precision, comprehensively analyzing segmentation quality.

4.3. Comparison with advanced technology

To demonstrate the effectiveness of the proposed hybrid network, we conducted several experiments on both the MAMA-MIA datasets and our in-house datasets. Accordingly, we compared the proposed method with various segmentation techniques, including nnFormer [55], PA-Net [56], PHTrans [57], nnUNet, PLHN, and STU-Net [58]. Table 1 presents the quantitative experimental results on the MAMA-MIA datasets. Evidently, IEDHTrans achieved the highest Dice coefficient (81.22%), which is 4.12% higher than the second-best STU-Net (77.10%). Table 2 presents the quantitative experimental results on the PLHN datasets, where IEDHTrans obtained a dice coefficient of 77.85%, which is 4.81% higher than the second-best STU-Net (73.04%). Similarly, Table 3 presents the quantitative comparison on the internal datasets, where IEDHTrans obtained a dice coefficient of 81.83%, which is 3.58% higher than the second-best STU-Net (78.25%). To further illustrate the effectiveness of the proposed method, qualitative experimental results are provided in Fig. 3, Figs. 4 and 5. Fig. 3 illustrates the segmentation performance of IEDHTrans on the MAMA-MIA dataset, Fig. 4 its performance on the PLHN dataset, while Fig. 5 shows its performance on our in-house dataset.

The quantitative comparisons in Tables 1–3 reveal the following phenomena: (1) Compared with other models on MAMA-MIA datasets, the proposed IEDHTrans achieves higher scores in most metrics, indicating that it generates segmentation results closer to the ground truth. (2) Compared with other models on our in-house dataset, the scores of the proposed IEDHTrans in various indicators are significantly higher than those of other models, demonstrating its clinical practical value. (3) IEDHTrans exhibited stable performance in the breast tumor segmentation task, indicating a high generalization ability. In summary, the proposed IEDHTrans demonstrates remarkable improvements in quantitative evaluations.

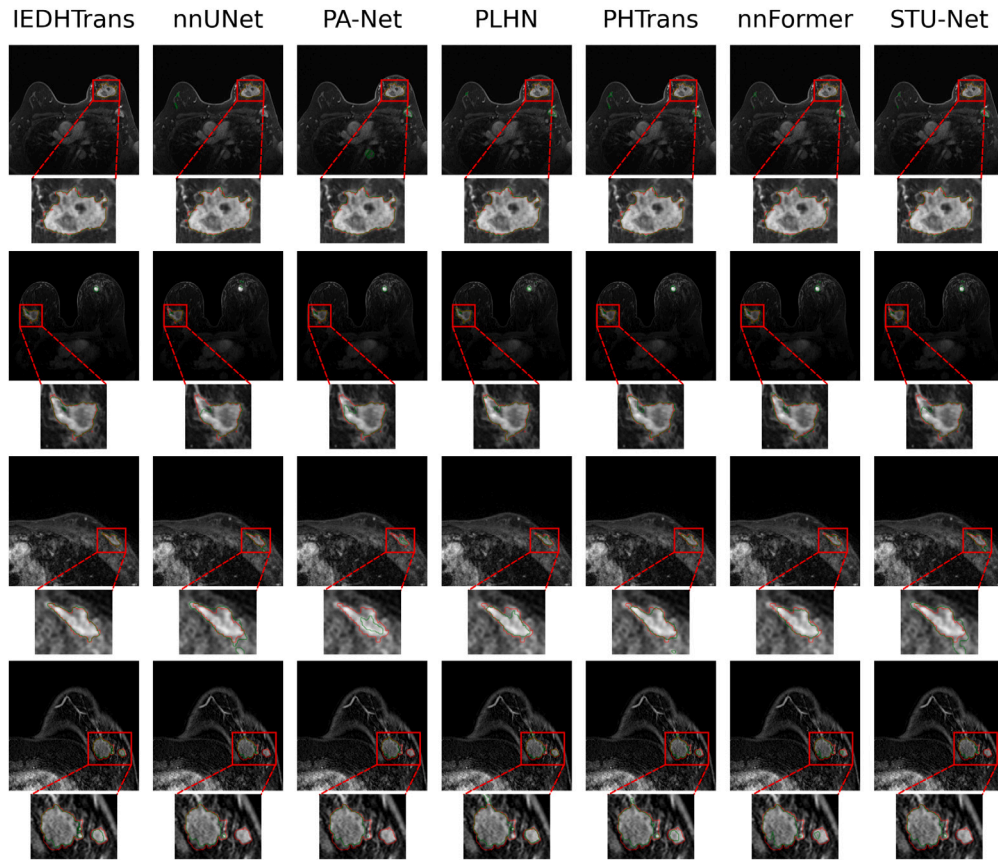


Fig. 3. Qualitative comparison of IEDHTrans with other methods on MAMA-MIA datasets. The region outlined by the red contour line represents the ground truth image and the region outlined by the green contour line is the predicted image generated by the model.

Table 3

Comparison of IEDHTrans with other methods on the internal dataset. The values in bold and underlined values represent the optimal and second-best values, respectively. The data in this table is a combination of mean $\pm$ standard deviation.

	DPC(%) $\uparrow$	ASSD(mm) $\downarrow$	VOE(%) $\downarrow$	HD95(mm) $\downarrow$	Sensitivity(%) $\uparrow$	Specificity(%) $\uparrow$
nnFormer	68.27 $\pm$ 18.26	29.03 $\pm$ 24.56	45.55 $\pm$ 19.72	102.36 $\pm$ 64.77	89.36 $\pm$ 8.82	99.96 $\pm$ 0.03
PA-Net	76.96 $\pm$ 20.24	18.11 $\pm$ 23.67	33.64 $\pm$ 24.08	76.11 $\pm$ 71.82	91.07 $\pm$ 10.79	99.97 $\pm$ 0.05
PHTrans	66.18 $\pm$ 23.25	37.95 $\pm$ 39.17	46.47 $\pm$ 24.47	199.82 $\pm$ 77.63	93.43 $\pm$ 7.22	99.94 $\pm$ 0.06
nnUNet	75.69 $\pm$ 20.94	<u>16.35 $\pm$ 27.53</u>	35.25 $\pm$ 23.85	<u>68.49 $\pm$ 70.19</u>	89.88 $\pm$ 10.74	99.97 $\pm$ 0.05
PLHN	67.24 $\pm$ 24.89	40.99 $\pm$ 41.05	44.65 $\pm$ 26.11	121.47 $\pm$ 64.22	80.55 $\pm$ 15.57	99.96 $\pm$ 0.06
STU-Net	<u>78.25 $\pm$ 20.84</u>	21.27 $\pm$ 26.75	<u>31.71 $\pm$ 24.38</u>	85.81 $\pm$ 77.43	89.75 $\pm$ 13.20	99.97 $\pm$ 0.04
IEDHTrans	81.83 $\pm$ 19.59	10.81 $\pm$ 22.06	28.59 $\pm$ 22.66	40.68 $\pm$ 57.39	95.61 $\pm$ 6.20	<u>99.97 $\pm$ 0.05</u>

The qualitative comparisons in Figs. 3–5, reveal the following phenomena: (1) The segmented images generated by the proposed IEDHTrans are closer to the real labeled images than the segmented images generated by the comparison model, both in qualitative external and internal dataset comparisons. (2) In both large and small tumor segmentation tasks, the segmented images generated by IEDHTrans appeared more convincing than the comparison. For example, the corresponding tumor regions of the segmentation images generated by IEDHTrans are all sharper and more realistic than the segmentation images of the comparison model. In summary, the proposed IEDHTrans exhibited high accuracy in qualitative evaluation.

We found that PLHN failed to reproduce the performance reported in its original paper when evaluated on its own publicly available dataset. We attribute this discrepancy to two main factors. First, the training data employed in PLHN's experimental setup differed from ours, although both studies used the PLHN dataset as an external test set. Second, PLHN relies on breast region masks during prediction,

whereas sufficient masks were not available in our setting to meet this requirement.

To provide a more intuitive illustration of the proposed IEDHTrans segmentation effectiveness, we plotted a box plot of six metrics on our internal dataset. This plot visually confirms the segmentation accuracy of IEDHTrans on each data point.

Fig. 6 shows that IEDHTrans performs exceptionally well in all six indicators and exhibits remarkable comprehensive performance. Its ASSD and HD95 values are relatively low, indicating that its segmentation boundary is precise and closely adheres to the true boundary. Its DPC and sensitivity values are relatively high, indicating that the target area is accurately identified, with a high degree of overlap and an ability to effectively capture fine structures. Its specificity value is also relatively high, reflecting a low misjudgment rate in non-target areas. VOE is relatively low, further verifying that the similarity between its segmentation result and real situation is high. Overall, IEDHTrans exhibits higher accuracy, enhanced robustness and more reliable performance than other models in the segmentation task.

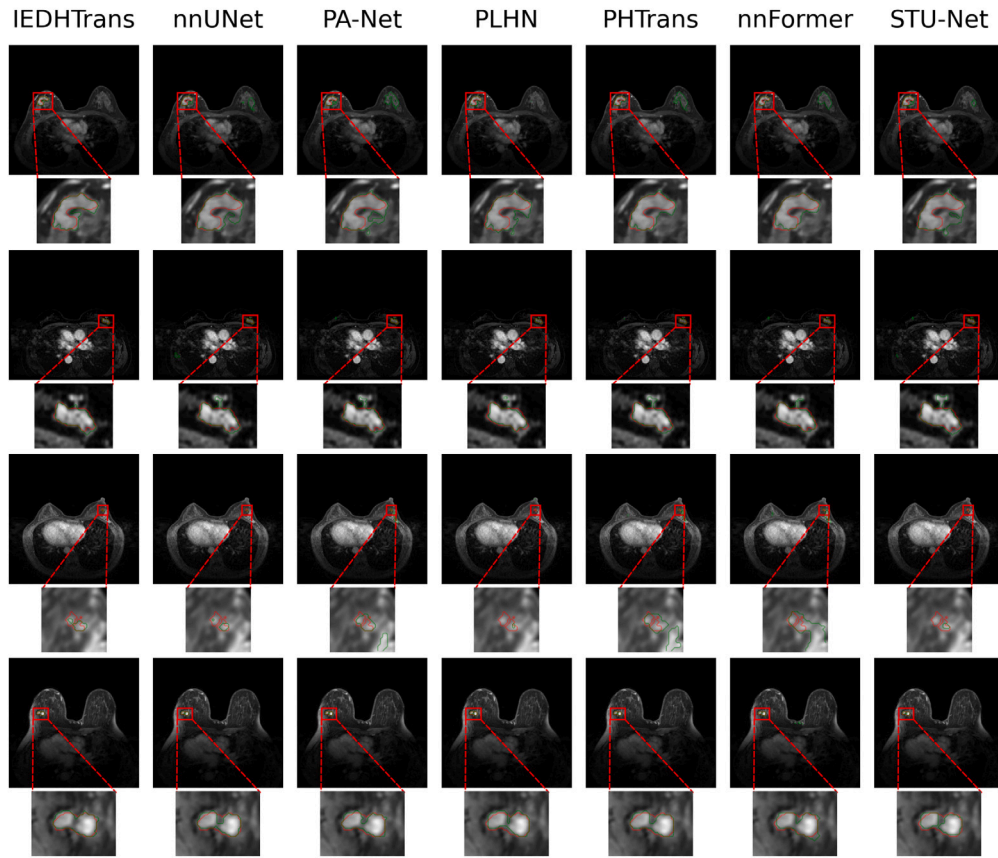


Fig. 4. Qualitative comparison of IEDHTrans with other methods on PLHN datasets. The region outlined by the red contour line represents the ground truth image and the region outlined by the green contour line is the predicted image generated by the model.

Table 4
T-test of IEDHTrans and six other methods on the Dice coefficient.

	MAMA-MIA dataset		PLHN dataset		internal dataset	
	t value	p value	t value	p value	t value	p value
nnFormer	7.9938	6.78e-15	5.2637	3.66e-07	2.8189	0.006519
nnUNet	3.2528	0.001207	2.4203	0.01641	1.1927	0.2377
PA-Net	3.8353	0.0001387	3.8435	0.0001633	0.9637	0.3391
PHTrans	3.7323	0.0002079	3.7148	0.0002645	2.8671	0.005708
PLHN	4.3508	1.60e-05	4.6895	5.10e-06	2.5644	0.01286
STU-Net	2.6431	0.00843	1.800	0.07338	0.6975	0.4882

To further demonstrate the advantages of IEDHTrans, we conducted a t-test between the dice coefficients of the remaining six models and IEDHTrans on the three datasets we used. The test results are shown in Table 4 as follows.

From Table 4, we can see that IEDHTrans maintains a statistically and clinically significant advantage over all mainstream models on large-scale data (MAMA-MIA, PLHN). When the data (our internal dataset) drops sharply, significance is diluted, which also indicates that the model's advantages are strongly coupled with the sample size. Subsequently, we will prioritize improving the model's performance in scenarios where data is scarce.

4.4. Ablation experiment

4.4.1. Multi-phase ablation

Herein, we perform ablation experiments on the MAMA-MIA datasets to demonstrate the effectiveness of the proposed multi-phase approach. First, we conducted experiments on the uniphase nnUNet, and the obtained results are presented in Table 5. The first row shows

that nnUNet model was trained using only pre-contrast images, while the second row was trained using only first post-contrast images. As presented in Table 5, by extracting and interacting with features from both pre- and first post-contrast images, IEDHTrans can better utilize multi-phase images to improve segmentation accuracy. Fig. 7 presents the qualitative experimental results, where the segmented image of IEDHTrans is closer to the real label image than that of the nnUNet model using only single modality.

4.4.2. Key component ablation

Furthermore, we conducted ablation experiments on the MPFI and DHFormer modules. Table 6 reports the quantitative results. The first row corresponds to the configuration with both MPFI and DHFormer removed. Rows two to four exclude the DHFormer module: row two presents the result of MPFI performing simple serial fusion, row three shows MPFI using only CBAM fusion, and row four corresponds to the complete MPFI module. Row five reports the performance of DHFormer when differential attention is replaced with self-attention, while row six shows the result when differential attention is replaced with window

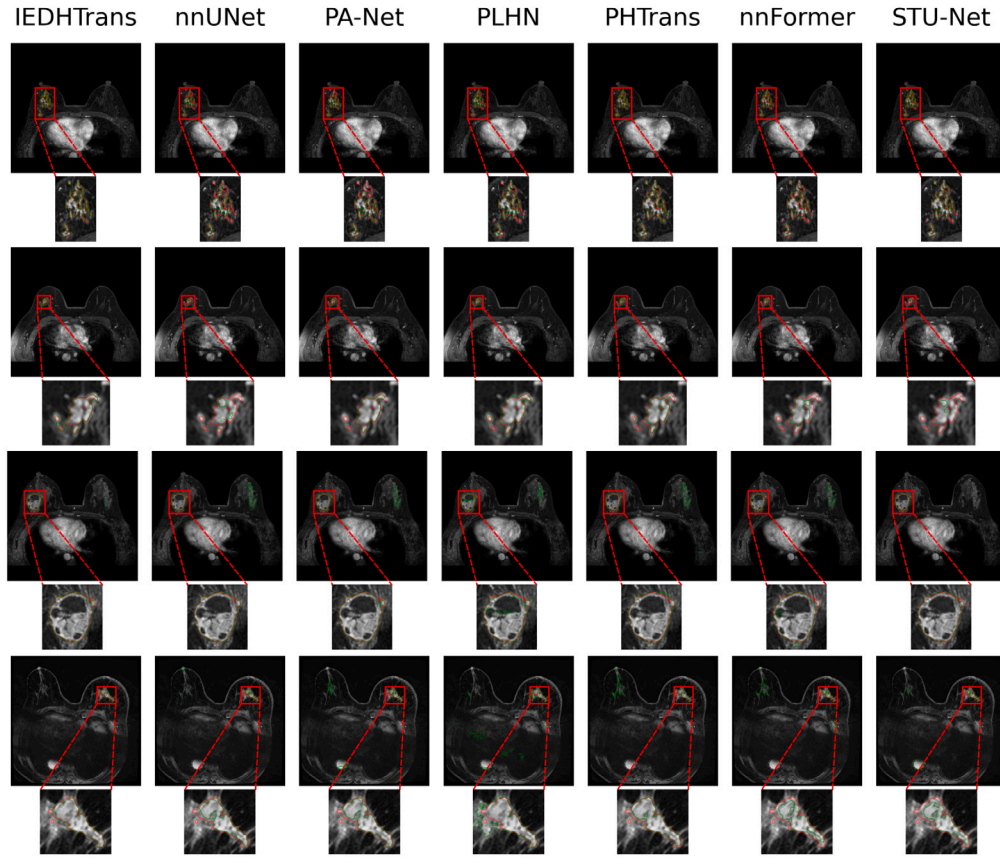


Fig. 5. Qualitative comparison of IEDHTrans with other methods on an internal dataset. The region outlined by the red contour line represents the ground truth image and the region outlined by the green contour line is the predicted image generated by the model.

Table 5

Comparison of IEDHTrans with single-modality approach. The values in bold are the optimal values. The data in this table is a combination of mean $\pm$ standard deviation.

	DPC(%) $\uparrow$	ASSD(mm) $\downarrow$	VOE(%) $\downarrow$	HD95(mm) $\downarrow$	Sensitivity(%) $\uparrow$
Pre-Contrast	60.82 $\pm$ 24.02	21.67 $\pm$ 29.58	52.21 $\pm$ 24.02	73.03 $\pm$ 82.00	72.57 $\pm$ 25.47
First Post-Contrast	73.71 $\pm$ 22.99	21.28 $\pm$ 36.62	37.11 $\pm$ 25.18	65.18 $\pm$ 83.36	82.84 $\pm$ 17.92
IEDHTrans	81.22 $\pm$ 16.75	6.99 $\pm$ 15.06	29.06 $\pm$ 20.14	30.39 $\pm$ 58.34	84.64 $\pm$ 18.03

Table 6

Experimental results of IEDHTrans after removing the key components. The values in bold are the optimal values. The data in this table is a combination of mean $\pm$ standard deviation.

	DPC(%) $\uparrow$	ASSD(mm) $\downarrow$	VOE(%) $\downarrow$	HD95(mm) $\downarrow$	Sensitivity(%) $\uparrow$
baseline	75.54 $\pm$ 21.35	15.85 $\pm$ 27.82	35.28 $\pm$ 23.80	58.21 $\pm$ 80.38	82.40 $\pm$ 19.18
baseline+Res	76.36 $\pm$ 20.05	14.18 $\pm$ 25.69	35.02 $\pm$ 23.60	51.72 $\pm$ 75.44	82.67 $\pm$ 17.55
baseline+CBAM	77.81 $\pm$ 18.87	10.33 $\pm$ 21.09	32.84 $\pm$ 21.19	40.41 $\pm$ 61.37	84.11 $\pm$ 19.22
baseline+MPFI	78.89 $\pm$ 18.39	9.86 $\pm$ 19.78	31.66 $\pm$ 21.35	39.41 $\pm$ 65.70	82.75 $\pm$ 18.67
baseline+MPFI+SHFormer	80.46 $\pm$ 17.31	8.37 $\pm$ 18.38	30.16 $\pm$ 20.47	32.62 $\pm$ 61.07	83.77 $\pm$ 18.07
baseline+MPFI+WHFormer	80.32 $\pm$ 17.76	11.01 $\pm$ 22.86	31.12 $\pm$ 21.98	40.06 $\pm$ 68.10	84.89 $\pm$ 17.43
baseline+MPFI+DHFormer	81.22 $\pm$ 16.75	6.99 $\pm$ 15.06	29.06 $\pm$ 20.14	30.39 $\pm$ 58.34	84.64 $\pm$ 18.03

attention. The final row represents the complete IEDHTrans. From Table 6, it can be observed that the CAGE block within MPFI facilitates higher-quality feature interaction through cross-attention, while CBAM proves effective for fusing 3D multiphase data. In addition, the differential attention mechanism in DHFormer enables more precise modeling of inter-phase feature differences via differential operations. Qualitative results are presented in Fig. 8 where IEDHTrans with its

key components achieves higher segmentation accuracy compared to versions with these components ablated.

5. Discussion

The proposed IEDHTrans method performs well in breast tumor segmentation tasks; however, it is necessary to acknowledge the limitations of this work. The collection and labeling of multi-modal breast

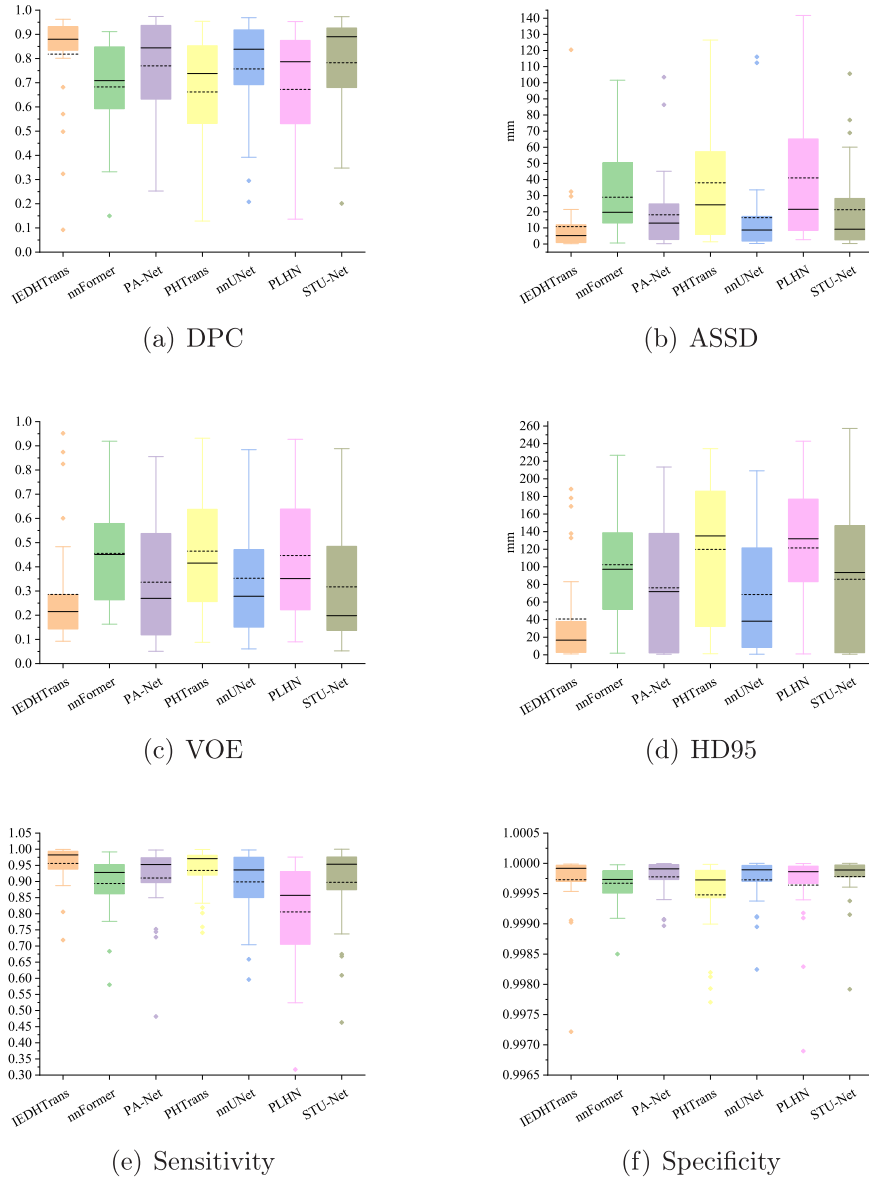


Fig. 6. Box plot of the six indicators on our internal dataset. Among them, the solid line represents the average value, the dashed line represents the median, the rhombus points represent outliers, and the range indicated by the box is from 25% to 75%.

cancer DCE-MRI data is significantly time-consuming and laborious. Our data are limited and data may be required from multiple medical centers to further validate the model. Nevertheless, IEDHTrans's results on both MAMA-MIA and internal datasets outperformed those of existing methods, demonstrating its beneficial role in clinical diagnosis and medical research. Future research directions should include expanding datasets, optimizing model structures and algorithms, and exploring integrations with other technologies to further improve performance and application range.

5.1. Performance of IEDHTrans

By integrating CNN and transformer benefits, IEDHTrans achieves significant performance gains in multi-modal breast cancer segmentation tasks. IEDHTrans's experimental results on both MAMA-MIA and internal datasets are better than current state-of-the-art methods. This indicates that IEDHTrans has high robustness and accuracy in processing complex medical images.

5.2. Effectiveness of key modules

IE module: The IE module extracts features from pre- and first post-contrast images via a parallel 3D CNN encoder that interacts with different phase features, significantly improving the fusion effect of multi-stage data. Experimental results demonstrate that the IE module can effectively capture the local features of breast tumors, thus improving segmentation accuracy. **DHFormer module:** The DHFormer module addresses the limitation of conventional CNN in modeling long-range pixel relationship by establishing global dependency between feature maps with different resolutions. The experimental results indicate that the proposed DHFormer module can significantly improve segmentation accuracy via its outstanding performance in multi-scale feature fusion.

5.3. Rationale for dual-phase image selection

Malignant breast tumors exhibit a distinctive contrast enhancement pattern. Specifically, early enhancement is pronounced, and the resulting time-signal intensity curves typically demonstrate either a

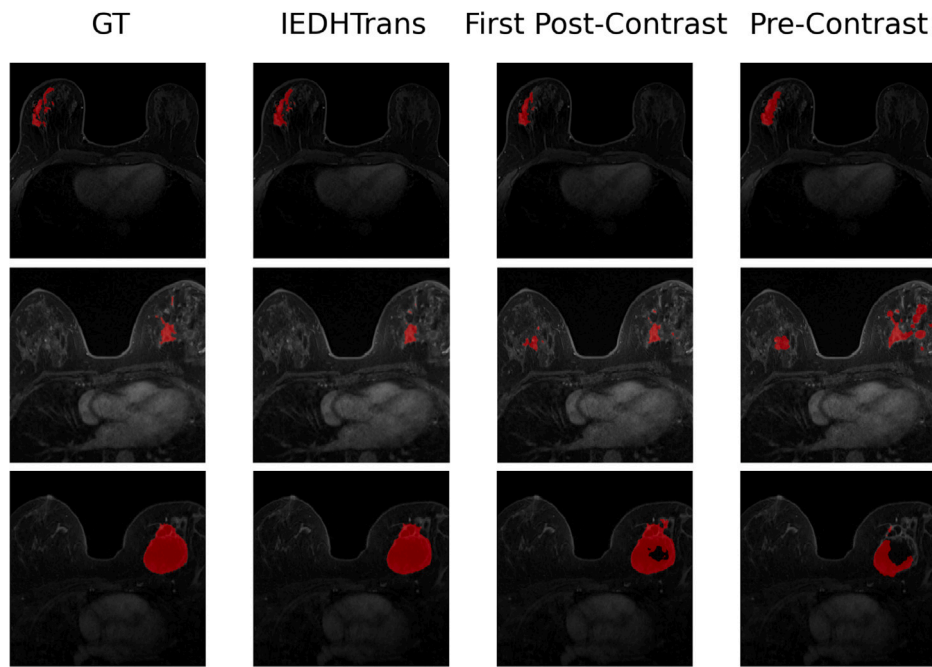


Fig. 7. Qualitative comparison of IEDHTrans with single-modality approach. The region outlined by the red contour line represents the ground truth image and the region outlined by the green contour line is the predicted image generated by the model. GT stands for ground truth.

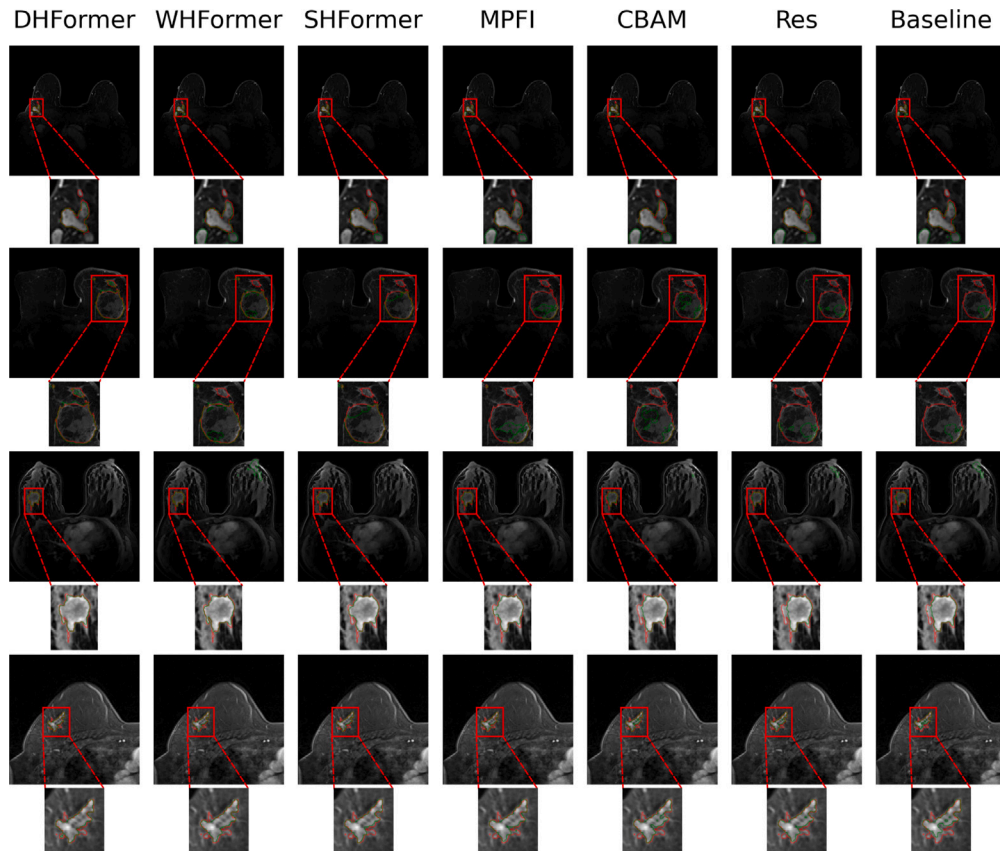


Fig. 8. Qualitative experimental results of IEDHTrans after removing the key components. The order from left to right in the figure corresponds to the sequence from bottom to top in Table 6.

rapid rise followed by a plateau or rapid rise followed by a slow decline. The lesion generally reaches its peak during the second to third enhancement phases, after which, no further enhancement is observed;

however, the enhancement level either stabilizes or decreases. Therefore, selecting the first post-contrast image is sufficient for assessing the tumor's progression in most cases.

5.4. Current challenges and future directions

The collection and labeling of current multi-phase breast cancer DCE-MRI data is time consuming and labor intensive. Future studies could expand the number of datasets to further validate the model's generalizability by including data from more medical centers. Although IEDHTrans performs optimally in the current task, there is still room for optimizing its computational complexity, and more lightweight model designs can be explored in the future to adapt to resource-constrained clinical settings. Concurrently, IEDHTrans can also be integrated with other advanced technologies (such as comparative learning, attention mechanisms, etc.) to further improve the model's performance and application range.

6. Conclusions

This study proposes IEDHTrans, a novel hybrid network that integrates the strengths of CNN and transformer networks for multi-phase breast cancer segmentation. Experimental results on the MAMA-MIA, PLHN and in-house clinical datasets confirm that the proposed IEDHTrans outperforms existing breast tumor segmentation methods. Its success can be primarily attributed to two key components: the IE and DHFormer modules. The IE module, a parallel 3D CNN encoder, leverages MPFI to extract features from pre- and first post-contrast images, significantly enhancing multi-phase data fusion. Conversely, the DHFormer module addresses the limitation of conventional CNNs in capturing long-range pixel dependencies by establishing global interactions among feature maps at different resolutions, which substantially improves segmentation accuracy. We suggest that IEDHTrans offers a substantial advancement for more accurate and efficient breast cancer diagnosis, providing a robust tool for researchers and clinicians alike. Future work will focus on expanding training dataset diversity and further refining the network architecture to enhance performance, ensuring that IEDHTrans evolves to satisfy the demands of precision medicine in oncology.

CRedit authorship contribution statement

Yuexin Wang: Conceptualization, Data curation, Formal analysis Methodology, Software, Writing – original draft. **Gesheng Song:** Data curation, Formal analysis, Investigation, Methodology, Resources, Visualization. **Jian Zhang:** Data curation, Software. **Fangqing Wang:** Data curation, Resources, Validation, Visualization. **Haixing Cheng:** Data curation, Methodology. **Yudan Zhao:** Formal analysis, Investigation, Validation. **Peng Zhou:** Validation, Visualization. **Xu Qiao:** Conceptualization, Resources, Supervision, Writing – review & editing. **Wei Chen:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing.

Ethics approval and consent to participate

The study involving human subjects was reviewed and approved by the Ethics Committee of the First Affiliated Hospital of Shandong First Medical University (No.2024-S974). Given the retrospective nature of this study, the requirement for obtaining informed consent was waived. This study was performed in line with the principles of the Declaration of Helsinki.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The code of the proposed approach and datasets are publicly available at <https://github.com/WYX-gh/IEDHTrans>.

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